

1. Find the tens digit of 6^{6^6} .
2. Find the last two digits of $6^{40} + 8^{40}$.
3. How many numbers less than 1000 are relatively prime to 1000?
4. Find the number of positive integers $n < 2018$ such that $25^n + 9^n$ is divisible by 13.
5. Define a list of numbers with the following properties:
 - The first number of the list is a one-digit natural number.
 - Each number (since the second) is obtained by adding 9 to the number before in the list.
 - The number 2012 is in that list.

Find the first number of the list.

6. Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was $\$ \underline{A} \underline{B} \underline{B} \underline{B} \underline{A}$, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?
7. What is the smallest positive integer n for which precisely 6 of the integers $n, 2n, 3n, \dots, 1000n$ are perfect squares?
8. Find the remainder when

$$\sum_{x=1}^{2024} \sum_{y=1}^{2024} (xy)$$

is divided by 31.

9. Zoe and Kylie each choose six distinct single digit positive integers and take the product of their numbers. If Zoe's product is 28 greater than Kylie's product, what is the sum of Zoe's six numbers?
10. Determine the smallest positive integer m such that $m^2 + 7m + 89$ is a multiple of 77
11. Find the number of positive integers $n < 100$ such that $\gcd(n^2, 2023) \neq \gcd(n, 2023^2)$.
12. Call a pair of positive integers (m, n) *tweenies* if and only if their least common multiple is 30. What is the sum of all possible products of two positive integers that are *tweenies*?
13. Define $f(n)$ to be the sum of the positive integer factors of n . If $f(n) = 360$ and $f(3n) = 1170$, what is the smallest possible value of $f(2n)$?
14. Let a, b , and c be positive integers satisfying the following equations:

$$\frac{\text{lcm}(a, b)}{\text{gcd}(a, b)} = 60 \quad \text{and} \quad \frac{\text{lcm}(b, c)}{\text{gcd}(b, c)} = 75.$$

What is the sum of all possible values of $\frac{\text{lcm}(c, a)}{\text{gcd}(c, a)}$?